

2025 NOV PURE MATHEMATICS PAPER 2 SOLUTIONS
6042/2 ADVANCED LEVEL

PURE MATHEMATICS
PAPER 2
6042/2

3 hours

ONLINE REVISION

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INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the spaces provided.
 - Answer all questions in Section A and any five questions from Section B.
 - Numerical answers should be given to appropriate accuracy.
-

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question.
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SECTION A (40 MARKS)

Answer all questions in this section.

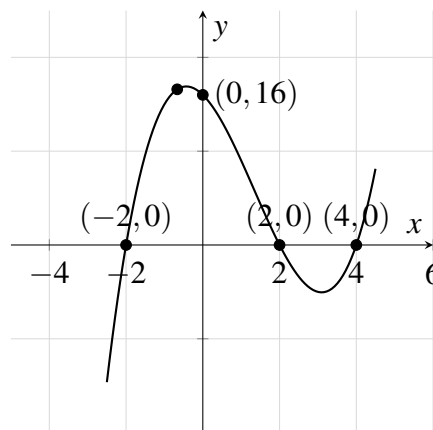
1. (a) Sketch on the same axes, the graphs of $y = |x^2 - 2|$ and $y = -x$. [3]
 (b) Hence or otherwise find the range of values of x for which

$$|x^2 - 2| < -x$$

.

 [3]

2. The diagram shows the graph of $y = f(x)$.



Sketch on separate diagrams, showing clearly all the intercepts, if any, the graphs of

- (a) $y = f(x) - 6$, [1]
 (b) $y = f(x + 1)$, [2]
 (c) $y = -\frac{1}{2}f(x)$, [2]
 (d) $y = f(2x)$. [2]
3. (a) By differentiating from first principles, find the gradient function of the function

$$f(x) = \frac{2x - 3}{x^2}$$

.

 [5]

- (b) Hence, or otherwise, find the equation of the tangent to the curve

$$f(x) = \frac{2x - 3}{x^2}$$

at the point where $x = 2$. [3]

4. The complex number u is given by $u = \frac{5+4i}{3-2i}$

(a) Express u in the form $x + iy$, where x and y are real numbers. [2]

(b) Hence or otherwise, find the modulus and argument of u . [4]

(c) Sketch an Argand diagram showing the point representing the complex number u . [2]

(d) Show on the same axes, the locus of the complex number z such that $|z - u| = 2$. [2]

5. The function $f : x \mapsto \frac{m}{x} + n$, where m and n are constants, is such that

$$f(-1) = 1\frac{1}{2} \quad \text{and} \quad f(2) = 9.$$

(a) State the domain of $f(x)$. [1]

(b) Find the value of m and n . [4]

(c) Evaluate $f^{-1}(4)$. [4]

SECTION B (80 MARKS)

Answer any five questions from this section. Each question carries 16 marks.

6. (a) Solve the inequality

$$\frac{2}{x-3} < \frac{3}{x-4}.$$

[5]

- (b) If $(1, 5)$ is a solution of the equations

$$n^2x + my = 14$$

$$ny - mx = 8$$

find the values of n and m .

[5]

- (c) i. Sketch on a single diagram the graphs of

$$y = 5 - |x| \quad \text{and} \quad y = |2x - 3|.$$

[3]

- ii. Hence, or otherwise, solve the inequality

$$|2x - 3| < 5 - |x|.$$

[3]

7. (a) i. Find the equation of the curve which satisfies the differential equation

$$\frac{dy}{dx} = \frac{-x}{y}$$

and passes through the point $(-4, 3)$.

[4]

- ii. Hence describe the solution curve.

[1]

- (b) The rate of destruction of drug by the kidney is proportional to the amount of the drug present in the body. At time t hours, the quantity of the drug in the body is x mg.

- i. Form a differential equation relating x and t .

[2]

- ii. Show that the general solution of the differential equation is

$$x = Ae^{-kt}$$

where A is a constant.

[3]

- iii. At $t = 0$, a quantity of 200 mg of the drug is administered. When $t = 1$ hour, the amount of the drug in the body is 180 mg. Find the quantity of the drug in the body after 8 hours. [6]

8. (a) The n^{th} term of a series is given by

$$4^{n-4} + 6n.$$

Find the sum of the first N terms. [6]

- (b) i. Find Maclaurin's series for

$$f(x) = (1 - x)^{-1}$$

up to and including the term in x^4 . [4]

- ii. Find the expansion of

$$(9 - x)^{\frac{3}{2}}(1 + 3x)^{\frac{3}{2}}$$

up to and including the term in x^2 . [5]

- iii. Hence find the set of values of x for which the expansion in (ii) is valid. [1]

9. (a) Express $2 \cos \theta - \sqrt{3} \sin \theta$ in the form $R \sin(\theta - \alpha)$. [3]

- (b) Solve the equation $2 \cos \theta - \sqrt{3} \sin \theta = 1$. [4]

- (c) Hence sketch the graph of $y = 2 \cos \theta - \sqrt{3} \sin \theta$, for $0^\circ \leq \theta \leq 360^\circ$. [4]

- (d) State the maximum and minimum values of $2 \cos \theta - \sqrt{3} \sin \theta$, for $0^\circ \leq \theta \leq 360^\circ$. [3]

- (e) Express

$$2 \cos^2 \theta - 1$$

in terms of $\cos 2\theta$. [2]

10. (a) Find an approximate value of

$$\int_0^1 x^2 e^x dx$$

using the trapezium rule with 5 ordinates. [5]

- (b) Evaluate the exact value of

$$\int_0^1 x^2 e^x dx$$

using integration by parts. [5]

- (c) Hence state whether the approximation obtained in (a) is an over-estimation or under-estimation. [1]

- (d) Use Newton-Raphson method to obtain a root of the equation

$$x^3 + 3x - 7 = 0$$

given that $x_0 = 1$. Give answer correct to 2 d.p. [5]

11. (a) An error of 3% is made in measuring the radius of a sphere. Find the percentage error in the volume. [volume of sphere = $\frac{4}{3}\pi r^3$]. [6]

- (b) i. Express

$$\frac{2x+1}{(x-2)(x^2+1)}$$

in partial fractions. [6]

- ii. Hence, show that

$$\int_0^1 \frac{2x+1}{(x-2)(x^2+1)} dx = -\frac{1}{2} \ln 8.$$

[4]

12. (a) Consider the table below for the set $R = (\{O, N, I, T\}r)$.

r	O	N	I	T
O	O	N	I	T
N	N	I	T	O
I	I	T	O	N
T	T	O	N	I

Show that $R = (\{O, N, I, T\}r)$ forms a group. [6]

- (b) i. Determine the single matrix M that describes an enlargement by factor 2 followed by a shear, factor 3 in the x-axis direction with y-axis invariant. [4]

- ii. Triangle ABC, with coordinates A(3,2), B(1,1) and C(5,1) is transformed using matrix M in (b)(i).

Find the coordinates of A' , B' and C' , the images of points A, B and C. [4]

- iii. Find the area of triangle $A'B'C'$. [2]

SOLUTIONS

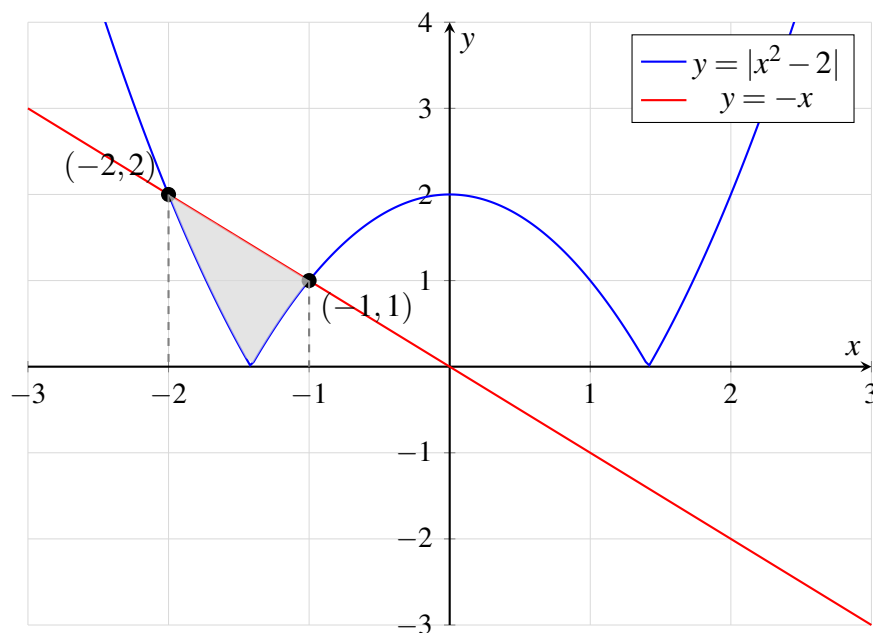
Section A

Question 1

Notes

Sketch both functions to understand their behavior, then find intersection points algebraically by considering different cases for the absolute value.

(a) Sketch:



(b) For $|x^2 - 2| < -x$, we need regions where the blue curve is below the red line.

Critical values where $|x^2 - 2| = -x$:

Case 1:

$$x^2 - 2 = -x$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \quad \text{or} \quad x = 1$$

Check: $x = -2$ valid ($-2 \leq -\sqrt{2}$), $x = 1$ invalid ($1 < \sqrt{2}$)

Case 2:

$$\begin{aligned} -(x^2 - 2) &= -x \\ x^2 - x - 2 &= 0 \\ (x - 2)(x + 1) &= 0 \\ x &= 2 \quad \text{or} \quad x = -1 \end{aligned}$$

Check: $x = 2$ invalid ($2 > \sqrt{2}$), $x = -1$ valid

Critical values: $x = -2$ and $x = -1$

$$-2 < x < -1$$

Question 2

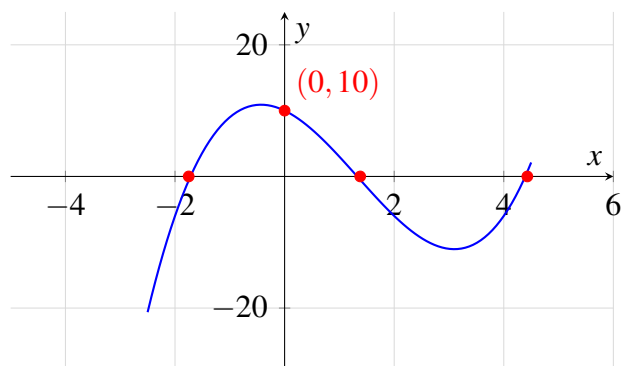
Notes

Apply vertical/horizontal translations, reflections, and stretches/compressions systematically to each key point. Given points from graph: $(-2, 4)$, $(0, 2)$, $(2, 0)$, $(4, 2)$

(a) $y = f(x) - 6$: Vertical translation downwards by 6 units

$$\begin{aligned} (-2, 4) &\rightarrow (-2, -2) \\ (0, 2) &\rightarrow (0, -4) \\ (2, 0) &\rightarrow (2, -6) \\ (4, 2) &\rightarrow (4, -4) \end{aligned}$$

$$y = f(x) - 6$$



(b) $y = f(x+1)$: (Horizontal translation left by 1 unit)

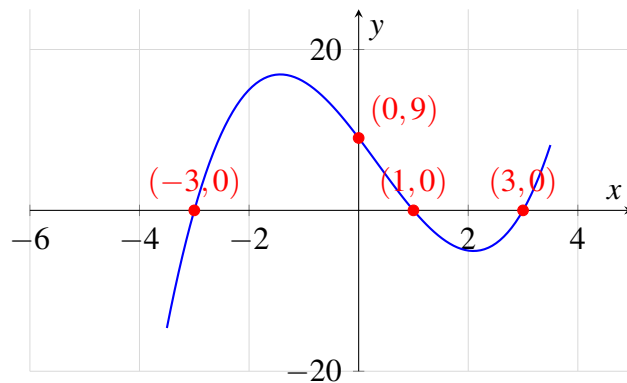
$$(-2, 4) \rightarrow (-3, 4)$$

$$(0, 2) \rightarrow (-1, 2)$$

$$(2, 0) \rightarrow (1, 0)$$

$$(4, 2) \rightarrow (3, 2)$$

$$y = f(x+1)$$



(c) $y = -\frac{1}{2}f(x)$: (Vertical reflection and compression by factor 1/2)

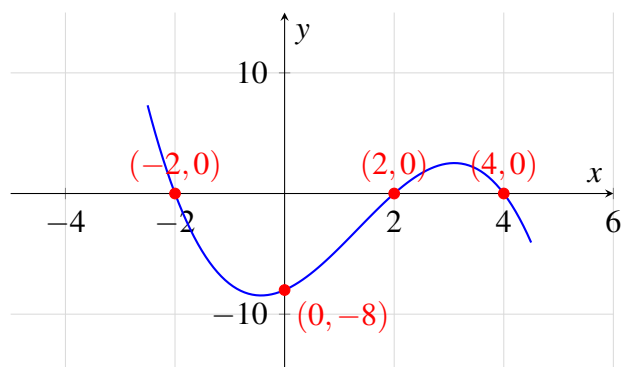
$$(-2, 4) \rightarrow (-2, -2)$$

$$(0, 2) \rightarrow (0, -1)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(4, 2) \rightarrow (4, -1)$$

$$y = -\frac{1}{2}f(x)$$



(d) $y = f(2x)$: (Horizontal compression by factor $1/2$)

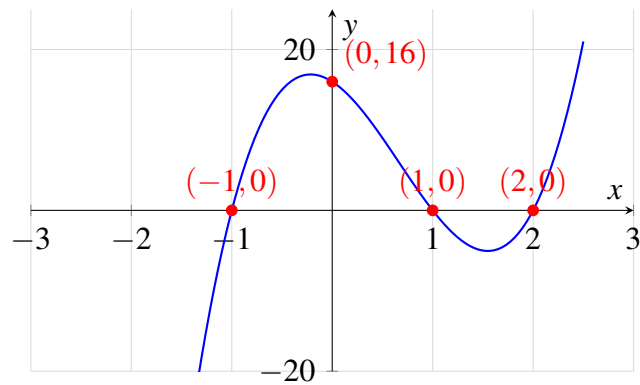
$$(-2, 4) \rightarrow (-1, 4)$$

$$(0, 2) \rightarrow (0, 2)$$

$$(2, 0) \rightarrow (1, 0)$$

$$(4, 2) \rightarrow (2, 2)$$

$$y = f(2x)$$



Question 3

Notes

Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ and simplify carefully.

(a)

$$\begin{aligned}
 f(x) &= \frac{2x-3}{x^2} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{2(x+h)-3}{(x+h)^2} - \frac{2x-3}{x^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^2[2(x+h)-3] - (2x-3)(x+h)^2}{hx^2(x+h)^2} \\
 &= \lim_{h \rightarrow 0} \frac{x^2(2x+2h-3) - (2x-3)(x^2+2xh+h^2)}{hx^2(x+h)^2} \\
 &= \lim_{h \rightarrow 0} \frac{2x^3+2x^2h-3x^2 - [2x^3+4x^2h+2xh^2-3x^2-6xh-3h^2]}{hx^2(x+h)^2} \\
 &= \lim_{h \rightarrow 0} \frac{-2x^2h-2xh^2+6xh+3h^2}{hx^2(x+h)^2} \\
 &= \lim_{h \rightarrow 0} \frac{-2x^2-2xh+6x+3h}{x^2(x+h)^2} \\
 &= \frac{-2x^2+6x}{x^4} = \frac{-2x+6}{x^3}
 \end{aligned}$$

Important Formula: First principles formula: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

(b) At $x = 2$:

$$\begin{aligned}
 f(2) &= \frac{2(2)-3}{2^2} = \frac{4-3}{4} = \frac{1}{4} \\
 f'(2) &= \frac{-2(2)+6}{2^3} = \frac{-4+6}{8} = \frac{2}{8} = \frac{1}{4}
 \end{aligned}$$

Equation of tangent:

$$\begin{aligned}
 y - \frac{1}{4} &= \frac{1}{4}(x-2) \\
 y &= \frac{1}{4}x - \frac{1}{2} + \frac{1}{4} \\
 y &= \frac{1}{4}x - \frac{1}{4} \\
 4y &= x - 1
 \end{aligned}$$

Question 4

Notes

Multiply numerator and denominator by conjugate to simplify, then find modulus and argument.

(a)

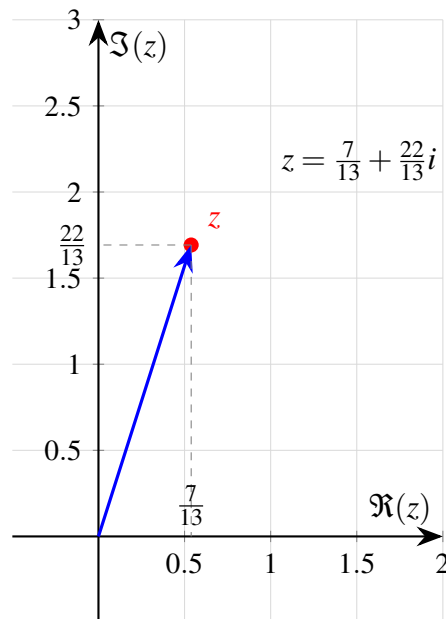
$$\begin{aligned}u &= \frac{5+4i}{3-2i} \times \frac{3+2i}{3+2i} \\&= \frac{(5+4i)(3+2i)}{9+4} \\&= \frac{15+10i+12i+8i^2}{13} \\&= \frac{15+22i-8}{13} \\&= \frac{7+22i}{13} \\&= \frac{7}{13} + \frac{22}{13}i\end{aligned}$$

(b) Modulus and argument:

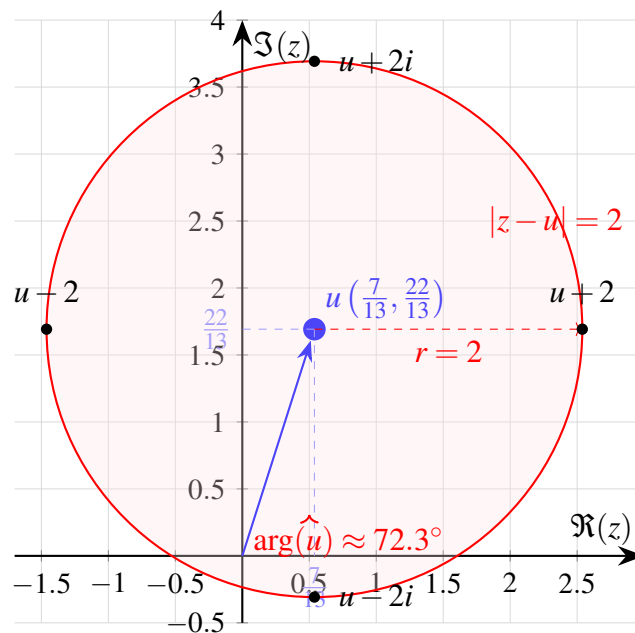
$$\begin{aligned}|u| &= \sqrt{\left(\frac{7}{13}\right)^2 + \left(\frac{22}{13}\right)^2} \\&= \frac{1}{13}\sqrt{49+484} = \frac{1}{13}\sqrt{533} \\&\approx 1.7759\end{aligned}$$

$$\begin{aligned}\arg(u) &= \tan^{-1}\left(\frac{22/13}{7/13}\right) \\&= \tan^{-1}\left(\frac{22}{7}\right) \\&\approx 72.3^\circ\end{aligned}$$

(c) Sketch;



(d) Argand diagram:



Question 5

Notes

Use given conditions to find m and n , then find inverse function algebraically.

(a) Domain: $x \in \mathbb{R}, x \neq 0$

(b) Given $f(x) = \frac{m}{x} + n$:

$$f(-1) = 1\frac{1}{2} = \frac{3}{2} :$$

$$f(2) = 9 :$$

$$\frac{m}{-1} + n = \frac{3}{2}$$

$$\frac{m}{2} + n = 9$$

Solving:

$$-m + n = \frac{3}{2} \quad (1)$$

$$\frac{m}{2} + n = 9 \quad (2)$$

From (1): $n = m + \frac{3}{2}$

Substitute into (2):

$$\frac{m}{2} + m + \frac{3}{2} = 9$$

$$\frac{3m}{2} + \frac{3}{2} = 9$$

$$3m + 3 = 18$$

$$3m = 15$$

$$m = 5$$

Then $n = 5 + \frac{3}{2} = \frac{13}{2}$

$$m = 5, \quad n = \frac{13}{2}$$

(c) Inverse function:

$$y = \frac{5}{x} + \frac{13}{2}$$

$$y - \frac{13}{2} = \frac{5}{x}$$

$$x = \frac{5}{y - \frac{13}{2}}$$

$$= \frac{5}{\frac{2y-13}{2}}$$

$$= \frac{10}{2y-13}$$

Thus $f^{-1}(x) = \frac{10}{2x-13}$

$$f^{-1}(4) = \frac{10}{2(4) - 13} = \frac{10}{8 - 13} = \frac{10}{-5} = -2$$

$$f^{-1}(4) = -2$$

Section B

Question 6

Notes

For rational inequality, combine fractions and analyze sign changes. For system, substitute given solution.

(a) Solve the inequality:

$$\begin{aligned}\frac{2}{x-3} &< \frac{3}{x-4} \\ \frac{2}{x-3} - \frac{3}{x-4} &< 0 \\ \frac{2(x-4) - 3(x-3)}{(x-3)(x-4)} &< 0 \\ \frac{2x-8-3x+9}{(x-3)(x-4)} &< 0 \\ \frac{1-x}{(x-3)(x-4)} &< 0\end{aligned}$$

Critical values: $x = 1$, $x = 3$, and $x = 4$

Sign analysis:

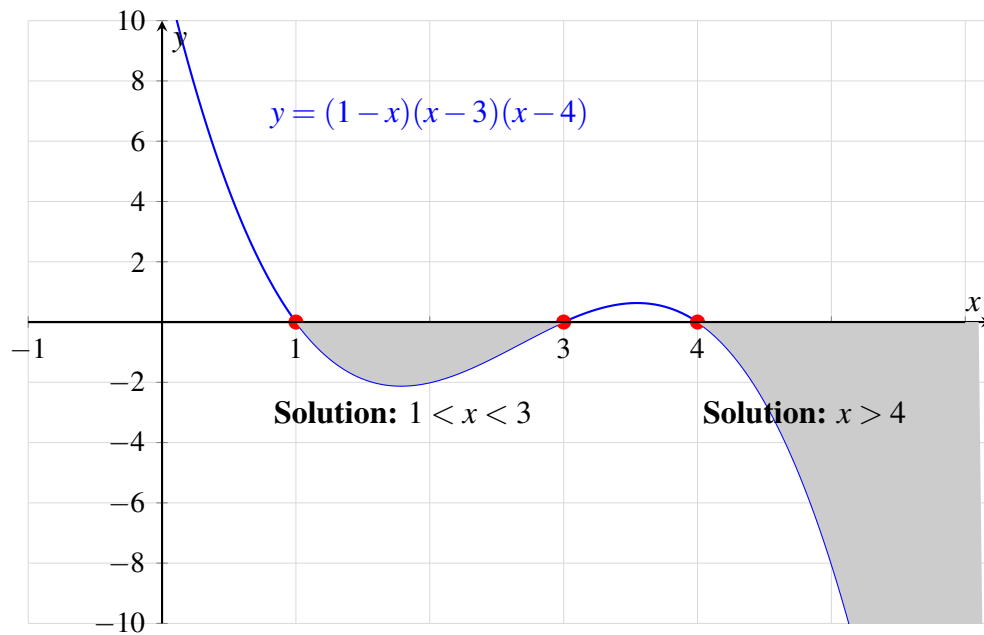
Interval	$(1-x)$	$(x-3)$	$(x-4)$	Product
$x < 1$	+	-	-	+
$1 < x < 3$	-	-	-	-
$3 < x < 4$	-	+	-	+
$x > 4$	-	+	+	-

The inequality $\frac{1-x}{(x-3)(x-4)} < 0$ is satisfied when the product is negative.

Graphical representation:

Graphical Interpretation

Consider the cubic function $y = (1-x)(x-3)(x-4)$ which has zeros at $x = 1$, $x = 3$, and $x = 4$. The inequality $\frac{1-x}{(x-3)(x-4)} < 0$ is equivalent to $(1-x)(x-3)(x-4) < 0$ (since the denominator squared is positive for $x \neq 3, 4$).



Therefore, the solution is:

$$1 < x < 3 \quad \text{or} \quad x > 4$$

Or

$$x \in (4, \infty) \cup (1, 3)$$

(b) Given $(1, 5)$ solves:

$$n^2(1) + m(5) = 14 \quad \Rightarrow \quad n^2 + 5m = 14 \quad (1)$$

$$n(5) - m(1) = 8 \quad \Rightarrow \quad 5n - m = 8 \quad (2)$$

From (2): $m = 5n - 8$

Substitute into (1):

$$n^2 + 5(5n - 8) = 14$$

$$n^2 + 25n - 40 = 14$$

$$n^2 + 25n - 54 = 0$$

$$(n + 27)(n - 2) = 0$$

$$n = -27 \quad \text{or} \quad n = 2$$

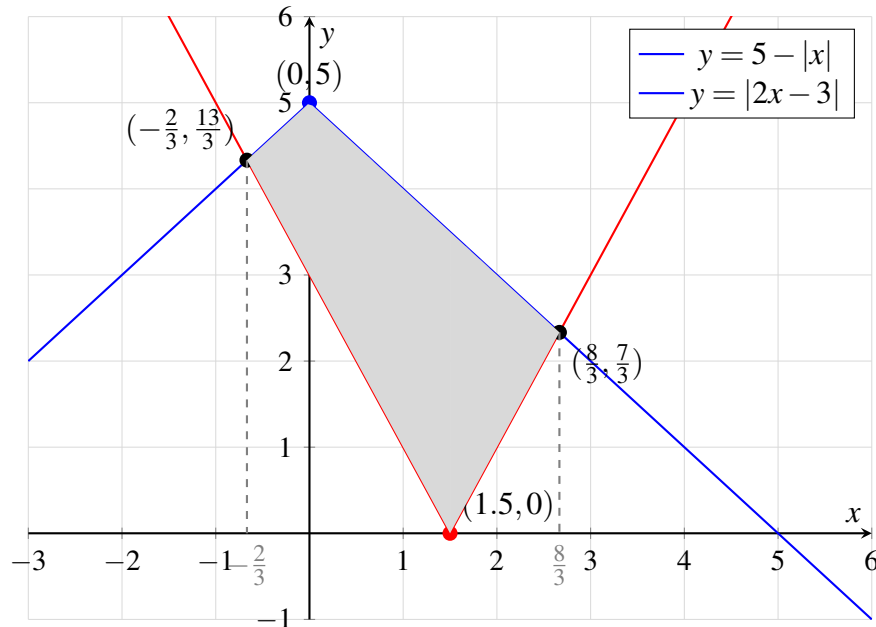
Corresponding m values:

$$\text{If } n = 2: \quad m = 5(2) - 8 = 2$$

$$\text{If } n = -27: \quad m = 5(-27) - 8 = -143$$

$$(m, n) = (2, 2) \text{ or } (-143, -27)$$

(c) (i) Graph:



(ii) For $|2x - 3| < 5 - |x|$:

Case 1: $x \leq 0$ (so $|x| = -x$, $|2x - 3| = -2x + 3$):

$$\begin{aligned} -2x + 3 &< 5 + x \\ -3x &< 2 \\ x &> -\frac{2}{3} \\ \text{Combine with } x \leq 0: & -\frac{2}{3} < x \leq 0 \end{aligned}$$

Case 2: $0 < x < \frac{3}{2}$ (so $|x| = x$, $|2x - 3| = -2x + 3$):

$$\begin{aligned} -2x + 3 &< 5 - x \\ -x &< 2 \\ x &> -2 \\ \text{Always true for } 0 < x &< \frac{3}{2} \end{aligned}$$

Case 3: $x \geq \frac{3}{2}$ (so $|x| = x$, $|2x - 3| = 2x - 3$):

$$2x - 3 < 5 - x$$

$$3x < 8$$

$$x < \frac{8}{3}$$

$$\text{Combine with } x \geq \frac{3}{2} : \quad \frac{3}{2} \leq x < \frac{8}{3}$$

Combining all cases:

$$-\frac{2}{3} < x < \frac{8}{3}$$

Question 7

Notes

Solve first DE by separating variables. The second models exponential decay.

(a) (i) Solve $\frac{dy}{dx} = -\frac{x}{y}$ through $(-4, 3)$:

$$y dy = -x dx$$

$$\int y dy = \int -x dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + C$$

$$y^2 + x^2 = 2C$$

$$\text{Using } (-4, 3): 9 + 16 = 2C \Rightarrow C = \frac{25}{2}$$

$$x^2 + y^2 = 25$$

(ii) The solution is a circle centered at origin with radius 5.

(b) (i) Rate $\propto$ amount: $\frac{dx}{dt} = -kx$ where $k > 0$

(ii) Solve:

$$\frac{dx}{dt} = -kx$$

$$\int \frac{1}{x} dx = \int -k dt$$

$$\ln x = -kt + C$$

$$x = e^{-kt+C} = Ae^{-kt} \quad (A = e^C)$$

(iii) Using conditions:

$$t = 0, x = 200 : 200 = Ae^0 \Rightarrow A = 200$$

$$t = 1, x = 180 : 180 = 200e^{-k}$$

$$e^{-k} = \frac{9}{10}$$

$$k = -\ln\left(\frac{9}{10}\right) = \ln\left(\frac{10}{9}\right)$$

After 8 hours:

$$\begin{aligned}x &= 200e^{-8k} = 200\left(e^{-k}\right)^8 \\&= 200\left(\frac{9}{10}\right)^8 \\&= 200 \times 0.43046721 \\&\approx 86.09 \text{ mg}\end{aligned}$$

After 8 hours: $\approx 86.09 \text{ mg}$

Question 8

Notes

Split series into geometric and arithmetic parts. Use binomial theorem for expansions.

$$(a) T_n = 4^{n-4} + 6n = \frac{4^n}{256} + 6n$$

Sum of first N terms:

$$\begin{aligned}S_N &= \sum_{n=1}^N T_n \\&= \frac{1}{256} \sum_{n=1}^N 4^n + 6 \sum_{n=1}^N n \\&= \frac{1}{256} \cdot \frac{4(4^N - 1)}{3} + 6 \cdot \frac{N(N+1)}{2} \\&= \frac{4^N - 1}{192} + 3N(N+1)\end{aligned}$$

$$S_N = \frac{4^N - 1}{192} + 3N(N+1)$$

(b) (i) Maclaurin for $(1-x)^{-1}$:

$$\begin{aligned}f(x) &= (1-x)^{-1} \\f'(x) &= (1-x)^{-2} \\f''(x) &= 2(1-x)^{-3} \\f'''(x) &= 6(1-x)^{-4} \\f^{(4)}(x) &= 24(1-x)^{-5}\end{aligned}$$

$$\text{At } x=0: f(0) = 1, f'(0) = 1, f''(0) = 2, f'''(0) = 6, f^{(4)}(0) = 24$$

$$\begin{aligned}f(x) &= 1 + x + \frac{2}{2!}x^2 + \frac{6}{3!}x^3 + \frac{24}{4!}x^4 + \dots \\&= 1 + x + x^2 + x^3 + x^4 + \dots\end{aligned}$$

(ii) Expand $(9-x)^{3/2}(1+3x)^{3/2}$:

First factor:

$$\begin{aligned}(9-x)^{3/2} &= 27 \left(1 - \frac{x}{9}\right)^{3/2} \\&= 27 \left[1 + \frac{3}{2} \left(-\frac{x}{9}\right) + \frac{\frac{3}{2} \cdot \frac{1}{2}}{2!} \left(-\frac{x}{9}\right)^2 + \dots\right] \\&= 27 \left(1 - \frac{x}{6} + \frac{x^2}{216} + \dots\right) \\&= 27 - \frac{9}{2}x + \frac{x^2}{8} + \dots\end{aligned}$$

Second factor:

$$\begin{aligned}(1+3x)^{3/2} &= 1 + \frac{3}{2}(3x) + \frac{\frac{3}{2} \cdot \frac{1}{2}}{2!}(3x)^2 + \dots \\&= 1 + \frac{9}{2}x + \frac{27}{8}x^2 + \dots\end{aligned}$$

Multiply:

$$\begin{aligned}
 & \left(27 - \frac{9}{2}x + \frac{x^2}{8}\right) \left(1 + \frac{9}{2}x + \frac{27}{8}x^2\right) \\
 &= 27 \cdot 1 + 27 \cdot \frac{9}{2}x + 27 \cdot \frac{27}{8}x^2 \\
 &\quad - \frac{9}{2}x \cdot 1 - \frac{9}{2}x \cdot \frac{9}{2}x + \frac{x^2}{8} \cdot 1 + \dots \\
 &= 27 + \frac{243}{2}x + \frac{729}{8}x^2 - \frac{9}{2}x - \frac{81}{4}x^2 + \frac{x^2}{8} + \dots \\
 &= 27 + \left(\frac{243}{2} - \frac{9}{2}\right)x + \left(\frac{729}{8} - \frac{162}{8} + \frac{1}{8}\right)x^2 + \dots \\
 &= 27 + 117x + \frac{568}{8}x^2 + \dots \\
 &= 27 + 117x + 71x^2 + \dots
 \end{aligned}$$

(iii) Validity: Need $|x/9| < 1$ and $|3x| < 1$, so $|x| < \frac{1}{3}$

$$-\frac{1}{3} < x < \frac{1}{3}$$

Question 9

Notes

Express in harmonic form $R \sin(\theta - \alpha)$, then solve and analyze.

(a) $2 \cos \theta - \sqrt{3} \sin \theta = R \sin(\theta - \alpha)$

Expanding: $R \sin(\theta - \alpha) = R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$

Equate coefficients:

$$R \cos \alpha = -\sqrt{3}$$

$$-R \sin \alpha = 2$$

$$\begin{aligned}
 R &= \sqrt{(-\sqrt{3})^2 + (-2)^2} = \sqrt{3+4} = \sqrt{7} \\
 \tan \alpha &= \frac{2}{\sqrt{3}} \Rightarrow \alpha = \tan^{-1} \left(\frac{2}{\sqrt{3}} \right) \approx 49.1^\circ
 \end{aligned}$$

Since $\cos \alpha < 0$ and $\sin \alpha < 0$, α in third quadrant: $\alpha \approx 180^\circ + 49.1^\circ = 229.1^\circ$

$$2 \cos \theta - \sqrt{3} \sin \theta = \sqrt{7} \sin(\theta - 229.1^\circ)$$

(b) Solve $\sqrt{7} \sin(\theta - 229.1^\circ) = 1$:

$$\sin(\theta - 229.1^\circ) = \frac{1}{\sqrt{7}} \approx 0.37796$$

$$\theta - 229.1^\circ = \sin^{-1}(0.37796) \quad \text{or} \quad 180^\circ - \sin^{-1}(0.37796)$$

$$\sin^{-1}(0.37796) \approx 22.2^\circ$$

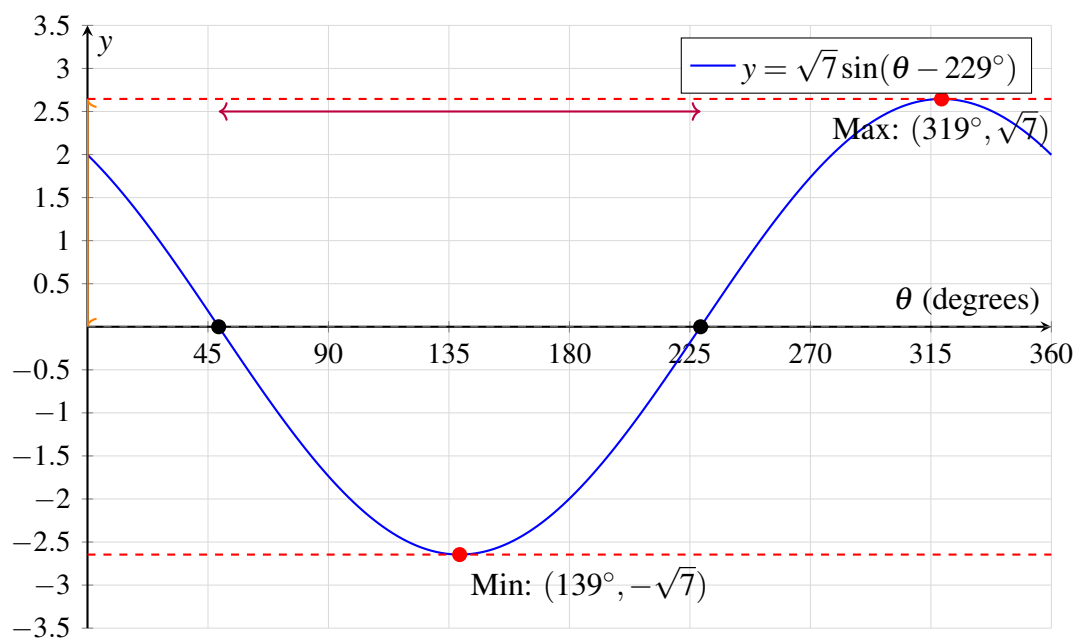
$$\theta \approx 229.1^\circ + 22.2^\circ \quad \text{or} \quad 229.1^\circ + 157.8^\circ$$

$$\approx 251.3^\circ \quad \text{or} \quad 386.9^\circ$$

$$\approx 251.3^\circ \quad \text{or} \quad 26.9^\circ$$

$$\theta \approx 27^\circ, 251^\circ (\text{to the nearest degree})$$

(c) Graph of $y = \sqrt{7} \sin(\theta - 229.1^\circ)$:



(d) Max value: $\sqrt{7}$, Min value: $-\sqrt{7}$

(e)

$$\begin{aligned} 2\cos^2 \theta - 1 &= 2\cos^2 \theta - (\cos^2 \theta + \sin^2 \theta) \\ &= 2\cos^2 \theta - \cos^2 \theta - \sin^2 \theta \\ &= \cos^2 \theta - \sin^2 \theta \\ &= \cos 2\theta \end{aligned}$$

Question 10

Notes

Apply trapezium rule and integration by parts. Use Newton-Raphson for root approximation.

(a) Trapezium rule with 5 ordinates ($n = 4$ strips):

$$h = \frac{1-0}{4} = 0.25$$

Values:

$$x_0 = 0 : y_0 = 0^2 e^0 = 0$$

$$x_1 = 0.25 : y_1 = 0.0625 e^{0.25} \approx 0.08025$$

$$x_2 = 0.5 : y_2 = 0.25 e^{0.5} \approx 0.41218$$

$$x_3 = 0.75 : y_3 = 0.5625 e^{0.75} \approx 1.19081$$

$$x_4 = 1 : y_4 = 1 \cdot e^1 \approx 2.71828$$

$$\begin{aligned} \int_0^1 x^2 e^x dx &\approx \frac{0.25}{2} [0 + 2.71828 + 2(0.08025 + 0.41218 + 1.19081)] \\ &= 0.125 [2.71828 + 2(1.68324)] \\ &= 0.125 [2.71828 + 3.36648] \\ &= 0.125 \times 6.08476 \approx 0.76060 \end{aligned}$$

Trapezium approx: 0.76 (2 d.p.)

(b) Integration by parts: Let $u = x^2 \implies du/dx = 2x$ and $dv/dx = e^x \implies v = e^x$

$$\begin{aligned} \int x^2 e^x dx &= x^2 e^x - \int 2x e^x dx \\ &= x^2 e^x - 2 \left(x e^x - \int e^x dx \right) \\ &= x^2 e^x - 2x e^x + 2e^x + C \end{aligned}$$

Evaluate from 0 to 1:

$$\begin{aligned} \int_0^1 x^2 e^x dx &= [x^2 e^x - 2x e^x + 2e^x]_0^1 \\ &= (e - 2e + 2e) - (0 - 0 + 2) \\ &= e - 2 \text{ (exact value)} \end{aligned}$$

(c) Trapezium rule (0.76) > exact (0.71828), so it's an **over-estimation**.

(d) Newton-Raphson for $f(x) = x^3 + 3x - 7$:

$$f'(x) = 3x^2 + 3, \quad x_0 = 1$$

$$x_1 = 1 - \frac{1^3 + 3(1) - 7}{3(1)^2 + 3} = 1 - \frac{-3}{6} = 1.5$$

$$x_2 = 1.5 - \frac{1.5^3 + 3(1.5) - 7}{3(1.5)^2 + 3} = 1.5 - \frac{0.875}{9.75} \approx 1.41026$$

$$x_3 = 1.41026 - \frac{f(1.41026)}{f'(1.41026)} \approx 1.41026 - \frac{0.019}{8.966} \approx 1.40814$$

$$x_4 \approx 1.40814$$

$$x \approx 1.41 \text{ (2 d.p.)}$$

Question 11

Notes

Use differentials for error analysis. For partial fractions, solve system of equations.

(a) Volume $V = \frac{4}{3}\pi r^3$, error $\delta r = 0.03r$

Using differentials:

$$\delta V \approx \frac{dV}{dr} \delta r = 4\pi r^2 \times 0.03r = 0.12\pi r^3$$

Percentage error:

$$\frac{\delta V}{V} \times 100\% = \frac{0.12\pi r^3}{\frac{4}{3}\pi r^3} \times 100\% = 0.12 \times \frac{3}{4} \times 100\% = 9\%$$

9%

Important Formula: Percentage error approximation: $\frac{\delta y}{y} \approx \frac{dy/dx \cdot \delta x}{y}$

(b) (i) Partial fractions:

$$\frac{2x+1}{(x-2)(x^2+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1}$$

Multiply through:

$$2x+1 = A(x^2+1) + (Bx+C)(x-2)$$

$$= (A + B)x^2 + (-2B + C)x + (A - 2C)$$

Equate coefficients:

$$A + B = 0$$

$$-2B + C = 2$$

$$A - 2C = 1$$

After solving simple algebra: $A = 1, B = -1, C = 0$

$$\frac{2x+1}{(x-2)(x^2+1)} = \frac{1}{x-2} - \frac{x}{x^2+1}$$

(ii) Integration:

$$\begin{aligned} \int_0^1 \frac{2x+1}{(x-2)(x^2+1)} dx &= \int_0^1 \left(\frac{1}{x-2} - \frac{x}{x^2+1} \right) dx \\ &= [\ln|x-2|]_0^1 - \frac{1}{2} [\ln(x^2+1)]_0^1 \\ &= (\ln 1 - \ln 2) - \frac{1}{2} (\ln 2 - \ln 1) \\ &= -\ln 2 - \frac{1}{2} \ln 2 \\ &= -\frac{3}{2} \ln 2 = -\frac{1}{2} \ln 8 \end{aligned}$$

Question 12

Notes

Verify group properties from Cayley table. For transformations, multiply matrices in correct order.

(a) Group verification:

- **Closure:** All table entries are in $\{O, N, I, T\}$
- **Identity:** O is identity ($a * O = O * a = a$)
- **Inverses:** $O^{-1} = O, N^{-1} = T, I^{-1} = I, T^{-1} = N$
- **Associativity:** $(O * N) * I = N * I = T$ and $O * (N * I) = O * T = T$, Hence $(O * N) * I = O * (N * I)$

Thus $(R, *)$ is a group (cyclic group of order 4).

(b) (i) Matrix M for enlargement then shear:

$$\text{Enlargement by 2: } E = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\text{Shear (x-direction, y-invariant): } S = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

$$\text{Combined: } M = S \cdot E = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix}$$

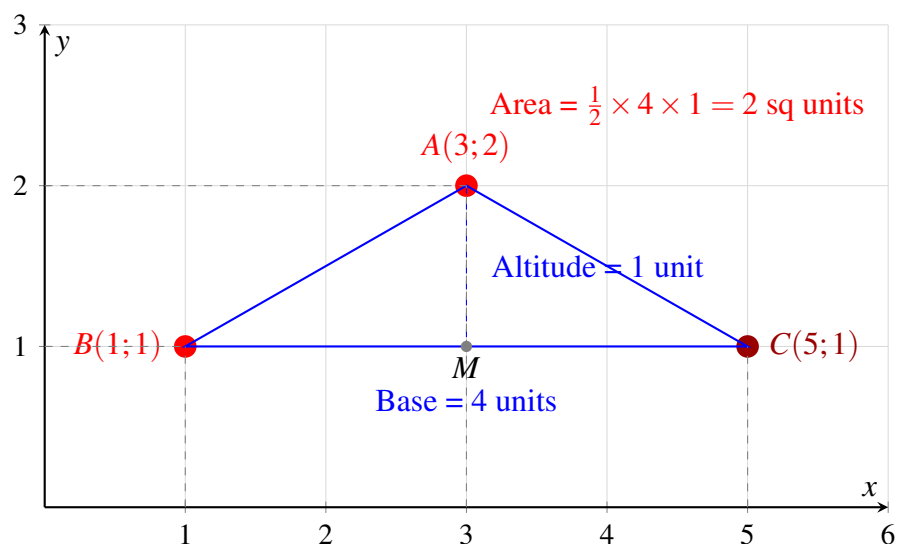
(ii) Transform triangle:

$$\text{Coordinates matrix: } \begin{pmatrix} 3 & 1 & 5 \\ 2 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 6 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 & 5 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 \cdot 3 + 6 \cdot 2 & 2 \cdot 1 + 6 \cdot 1 & 2 \cdot 5 + 6 \cdot 1 \\ 0 \cdot 3 + 2 \cdot 2 & 0 \cdot 1 + 2 \cdot 1 & 0 \cdot 5 + 2 \cdot 1 \end{pmatrix} = \begin{pmatrix} 18 & 8 & 16 \\ 4 & 2 & 2 \end{pmatrix}$$

$$A'(18,4), B'(8,2), C'(16,2)$$

(iii) Area transformation:



$$\det(M) = 2 \times 2 - 6 \times 0 = 4$$

Original triangle area (using determinant method):

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 3 & 1 & 5 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \frac{1}{2} |3(1-1) - 1(2-1) + 5(2-1)| = 2$$

$$\text{New area} = |\det(M)| \times \text{original area} = 4 \times 2 = 8$$

$$8 \text{ units}^2$$

END OF SOLUTIONS

**FEEL FREE TO CONTACT US FOR ANY ADJUSTMENTS, CLARIFICATIONS AND
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